



May 2025

EC3261-1 – ARTIFICIAL INTELLIGENCE

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

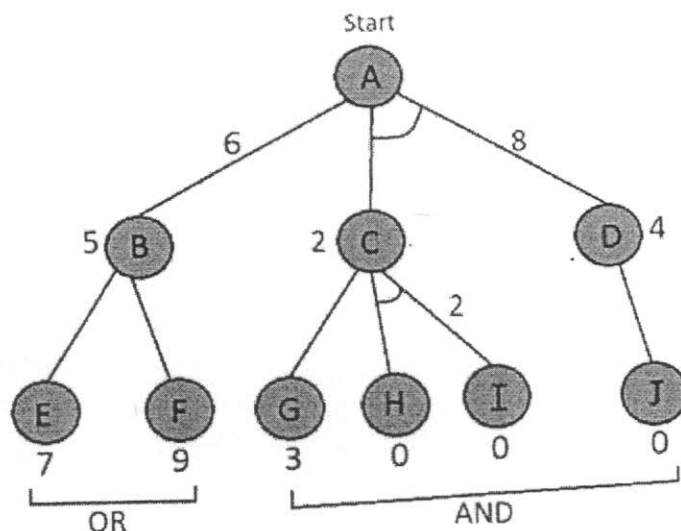
- The following is the ignorable problem**
A) Theorem proving
B) 8 puzzle
C) Chess
D) None of these
- The following is an example for partially commutative monotonic system**
A) Theorem proving
B) Robot Navigation
C) Chemical Synthesis
D) Bridge
- In Breadth-First Search algorithm, the time complexity is**
A) $O(b^{d+1})$
B) $O(b^d)$
C) $O(b^{d-1})$
D) $O(b^d + 2)$
- The following is an example for recoverable problem.**
A) Theorem proving
B) 8 puzzle
C) Chess
D) None of these
- In Natural Language Processing, search required to find a complete interpretation for the sentence is called**
A) a state of the world
B) problem a state of the world
C) solution a state of the word
D) solution a state of the world
- Finding a shortest path to a travelling salesman problem is the following problem characteristic**
A) Is the problem decomposable into a set of (nearly) independent smaller or easier sub problems?
B) Can solution steps be ignored or at least undone if they prove unwise?
C) Is the problem's universe predictable?
D) Is a good solution to the problem?
- AO* Search algorithm uses the following graph**
A) NOT graph
B) OR Graph
C) AND OR graph
D) NOR Graph
- The optimal solution in the state-space diagram is represented by the state where the objective function reaches its maximum value, also known as the _____**
A) Local maximum
B) Global maximum
C) shoulder
D) Ridge
- The ability to represent all of the kinds of knowledge that are needed in that domain is known as**
A) Inferential Adequacy
B) Inferential Efficiency
C) Acquisitional Efficiency
D) Representational Adequacy

10. The programming language LISP is used in _____
 A) Inferential Knowledge B) Procedural Knowledge
 C) Inheritable Knowledge D) Simple relational knowledge
11. _____ is a collection of attributes and associated values that describe some entity in the world.
 A) Signal B) Nets
 C) Frames D) Videos
12. _____ is a structure that describes a stereotyped sequence of events in a particular context
 A) Probability B) Linear Algebra
 C) Conceptual Dependency D) Script
13. The final step in pragmatic Analysis is to _____
 A) Check B) Translate
 C) hierarchical structure D) Integration
14. According to Chomsky hierarchy, grammar, Type 1 is known as
 A) Unrestricted Grammar B) Context-sensitive grammar
 C) Context free grammar D) Regular Grammar
15. _____ is a linguistic theory that analyses how verbs relate to their arguments in a sentence
 A) Case Free Grammar B) Context-sensitive grammar
 C) Transformational grammar D) Regular Grammar
16. Using Fourier Transform, image can be displayed as _____
 A) Color Function B) Intensity Function
 C) Length Function D) width function
17. The _____ of Fourier spectra is usually higher than the typically display device.
 A) Dynamic Range B) Static Range
 C) Color Range D) None of these
18. _____ is a process used to resize a digital image.
 A) image rotation B) image scaling
 C) image segmentation D) image reflection
19. The following is not an example for AI in telemarketing
 A) Analyze customer data B) Personalized interactions
 C) Optimize marketing strategies D) Image Analysis
20. _____ is used to divide large group of data points
 A) Segment correction B) Search
 C) Fourier Transform D) Segment analysis

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) A water jug problem states "you are provided with two jugs, first one with 4 gallon capacity and the second one with 3 gallon capacity. Neither have any measuring markers on it. How can you get exactly 2 gallons of water into 4-gallon jug? (i) Write down the production rules for the above problem (ii) Write any one solution to the above problem	8	L*3	1	1
	b) Discuss hill climbing algorithm with applications.	8	L3	1	1
2.	a) Discuss The Missionaries and Cannibals Problem	8	L3	1	1
	b) Discuss Issues in the Design of Search Programs.	8	L3	2	1

3. a) Apply AO* algorithm for the following problem.



- b) Discuss different components of AI.

8	L3	2	1
8	L3	2	1

Unit – II

4. a) What is a Script? Discuss the components of a script.
 b) Discuss Case Grammars with an example. Also, discuss the importance of case Grammar.
5. a) With a block diagram, discuss the Conceptual Dependency representation and its key components.
 b) Discuss different tasks involved in Natural Language Processing.
6. a) Discuss Semantic Network and its types in detail.
 b) Discuss any eight applications of NLP.

8	L3	3	1
8	L3	4	1
8	L3	4	1
8	L3	3	1
8	L3	3	1
8	L3	4	1

Unit – III

7. a) Discuss the working of template matching.
 b) How AI can be used to Telemarketing? Briefly, discuss with an example.
8. a) Discuss the different edge detection techniques available.
 b) Discuss the working of optical sensors.

8	L3	5	1
8	L3	5	1
8	L3	5	1
8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
